# Tingfeng Lan

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## Research Interests

**Keywords: LLM infra, storage systems for AI, serverless computing**

- The overarching goal of my research is to enable fundamentally sustainable, scalable, and performant AI applications and platforms by exploring the intersection of AI and systems.

- Current research focuses: (1) rethinking the LLM storage backend through new system abstractions that better align with modern model and workload characteristics, and (2) cost-effective training and inference systems on modern and heterogeneous hardware platforms.

## Education

**University of Virginia** Sep 2024 – Present
*Ph.D. in Computer Science, Advisor: Prof. Yue Cheng* *VA, USA*

**Sichuan University** Sep 2020 – Jun 2024
*B.Eng. in Computer Engineering, Advisor: Prof. Mingjie Tang* *Sichuan, China*

## Industry Experience

**AntGroup AI Infra** Sep 2023 – Jul 2024
Research Intern, Manager: Jian Sha

*- Designed and implemented **DLRover-RM** (VLDB'24), a resource-aware optimization system for large-scale recommendation-model training that improves resource utilization and reduces training cost in cloud environments.*

*- Designed and implemented **m-LoRA** (VLDB'25), a multi-tenant LoRA training framework that enables parallel multi-adapter fine-tuning via pipeline parallelism, reducing memory redundancy and improving training throughput.*

## Publications

**Preprint** Tingfeng Lan, Zirui Wang, Zhaoyuan Su, Yunjia Zheng, Juncheng Yang, Yue Cheng. "**TStore: Towards Practical Compression-Aware Model Storage with Tensor-Centric Fingerprinting and Clustering**."

**Preprint** Yinghao Tang, Tingfeng Lan, Xiuqi Huang, Hui Lu, Wei Chen. "**SCORPIO: Serving the Right Requests at the Right Time for Heterogeneous SLOs in LLM Inference**."
*In submission to IJCAI'26.*

**Preprint** Tingfeng Lan, Yusen Wu, Bin Ma, Zhaoyuan Su, Rui Yang, Tekin Bicer, Masahiro Tanaka, Olatunji Ruwase, Dong Li, Yue Cheng. "**ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates**."
*Adopted into DeepSpeed and featured on PyTorch blog.*

**ACL Demo'26** Yinghao Tang, Yupeng Xie, Yingchaojie Feng, Tingfeng Lan, Wei Chen. "**IGenBench: Benchmarking the Reliability of Text-to-Infographic Generation**."

**ACL'26** Yinghao Tang, Xueding Liu, Boyuan Zhang, Tingfeng Lan, Yupeng Xie, Jiale Lao, Yiyao Wang, Haoxuan Li, Tingting Gao, Bo Pan, Luoxuan Weng, Xiuqi Huang, Minfeng Zhu, Yingchaojie Feng, Yuyu Luo, Wei Chen. "**IGenBench: Benchmarking the Reliability of Text-to-Infographic Generation**."

**MLSys'26** Zhaoyuan Su, Tingfeng Lan, Zirui Wang, Juncheng Yang, Yue Cheng. "**Efficient and Workload-Aware LLM Serving via Runtime Layer Swapping and KV Cache Resizing**."

**MLSys'26** Minchen Yu, Rui Yang, Chaobo Jia, Zhaoyuan Su, Sheng Yao, Tingfeng Lan, Yuchen Yang, Yue Cheng, Wei Wang, Ao Wang, Ruichuan Chen. "**$\lambda$Scale: Enabling Fast Scaling for Serverless Large Language Model Inference**."

**Preprint** Jiale Lao, Yinghao Tang, Tingfeng Lan, Mingjie Tang, Yuanchuan Zhou, Jianguo Wang. "**PathBee: Accelerating Shortest Path Querying via Graph Neural Networks**."
*In submission to ICDE'26.*

**ICDE'26** Ziling Huang, Zhengmao Ye, Qingsong Cai, Zelong Huang, Bo Sang, Haitao Zhang, Jian Sha, Tingfeng Lan, Hui Lu, Yuanchun Zhou, Mingjie Tang. "**DLRover-LM: LLM Pre-Training Framework with Thousands of Accelerators in AntGroup**."
*In Proceedings of the 42nd IEEE International Conference on Data Engineering.*

**NSDI'26** Zirui Wang, Tingfeng Lan, Zhaoyuan Su, Juncheng Yang, Yue Cheng. "**ZipLLM: Efficient LLM Storage via Model-Aware Synergistic Data Deduplication and Compression**."
*In Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation.*

**VLDB'25** Zhengmao Ye*, Dengchun Li*, Zetao Hu, Tingfeng Lan, Jian Sha, Sicong Zhang, Lei Duan, Jie Zuo, Hui Lu, Yuanchun Zhou, Mingjie Tang. "**mLoRA: Fine-Tuning LoRA Adapters via Highly-Efficient Pipeline Parallelism in Multiple GPUs**."
*In Proceedings of 51$^{th}$ International Conference on Very Large Data Bases*

**VLDB'24** Qinglong Wang*, Tingfeng Lan*, Yinghao Tang, Bo Sang, Haitao Zhang, Jian Sha, Hui Lu, Ke Zhang, Mingjie Tang. "**DLRover-RM: Resource Optimization for Deep Recommendation Models Training in the Cloud**."
*In Proceedings of 50$^{th}$ International Conference on Very Large Data Bases*

* denotes equal contribution

## Open Source Projects

**DeepSpeed-ZenFlow: A stall-free offloading framework for LLM fine-tuning** Oct 2024 - Present
Available on DeepSpeed, Received 40k+ ⭐ on GitHub
*- Designed and implemented **ZenFlow**, an importance-aware asynchronous offloading system that decouples GPU and CPU updates to eliminate GPU stalls. Achieved up to 5× end-to-end speedup, 2× reduction in PCIe traffic, and over 85% stall elimination while preserving accuracy.*

**mLoRA: A efficient multi-tenant LoRA training system** Sep 2023 - May 2024

Received 300+ ⭐ on GitHub

*- Designed and implemented a training mechanism "BatchLoRA" which allows multiple LoRA adapters to share the pre-trained base model concurrently with reduced kernel launch overhead.*

**DLRover: An efficient autodl system with fault-tolerance awareness** Jun 2023 - March 2024

Received 1.6k+ ⭐ on GitHub, Joined LF AI & Data Foundation ⚡

*- Designed and implemented a hyper-parameter autotuner to optimize performance-relevant configurations, like micro-batch size, for maximum hardware utilization. Achieved over 95% memory utilization within a 30s estimation and re-configuration time; An elastic trainer, allowing for real-time hyper-parameter configuration during training sessions, thereby eliminating the restart overheads typically necessary in conventional training frameworks.*

## Funding and Grants

2025 **Modal Research Grant;** Total: $2,000 GPU credits; Led the proposal writing and submission;

## Service & Activities

2025-2026 **Shadow PC for EuroSys'26**

2025-2026 **Artifact Evaluation Committee for EuroSys'26, NSDI'26, OSDI'26**

2023–2024 **Journal Reviewer for IEEE TBD'24**

## Talks

2025 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates;**
Invited talk: DeepSpeed Team

2025 **ZenFlow: Enabling Stall-Free Offloading Training via Asynchronous Updates;**
Poster presentation: UVA 2025 AI/ML Resource Fair

## Awards

2026 **EuroSys Distinguished AEC**